

United International University

Department of Computer Science and Engineering

Data Mining Project Report

**Spatial-Temporal Demand Analysis and
Customer Sentiment Mining for
On-Demand Services**

Course

CSE 4891 – Data Mining

Section

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Faculty

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1 Introduction

The major goal of this project is to study the Yelp Open Dataset and to use a variety of data mining techniques to generate valuable spatiotemporal, predictive, and customer-oriented information. By analyzing both structured business information and unstructured customer reviews, the project aims to identify meaningful patterns related to business activity, geographical concentration, temporal behavior, and customer experiences. The analysis combines multiple data mining approaches so that the dataset can be examined from different perspectives rather than focusing on a single analytical task.

The project is based on a comprehensive data mining pipeline consisting of data collection, data preprocessing, exploratory data analysis, spatial clustering, demand forecasting, sentiment analysis, aspect-based analysis, and interactive visualization. The different stages are connected sequentially, where the processed output of one stage is used as input for subsequent analysis. Exploratory analysis is first performed to understand the characteristics and distribution of the data, followed by spatial and predictive techniques to identify geographical patterns and estimate future activity. Finally, natural language processing techniques are applied to extract customer-oriented information from review text, and the results are presented through an interactive dashboard.

The Yelp Open Dataset does not contain service requests, bookings, sales, or transaction records. Thus, this project employs review activity as an indicator of demand for service. That is, the number of reviews that are received for a business during a given month are treated as an observed measure of customer activity. This proxy-based approach allows the project to investigate temporal changes in customer activity even though direct transaction-level demand information is unavailable. Therefore, the forecasting results should be interpreted as predictions of review activity rather than direct predictions of actual service transactions.

The main predictive task of this project is monthly review-activity forecasting. Five regression models were trained on the historical activity features:

- Random Forest
- XGBoost
- Linear Regression
- Decision Tree
- HistGradientBoosting

The models were tested by the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 score.

Besides forecasting, DBSCAN clustering was applied to detect the business areas where it was geographically dense. For the purpose of customer sentiment analysis, VADER was utilized and to examine the three specific elements, namely Price, Punctuality and Service Quality, rule-based keyword matching was employed.

Lastly, the outcomes were combined into an interactive Streamlit dashboard.

2 Dataset Used

The project uses the Yelp Open Dataset, which contains information about businesses, reviews, users, tips, and check-ins.

The dataset was obtained from the publicly accessible Kaggle platform:

<https://www.kaggle.com/yelp-dataset/yelp-dataset>

The main files used in this project are:

- `yelp_academic_dataset_business.json`
- `yelp_academic_dataset_review.json`

The business dataset contains information such as:

- Business ID
- Business name
- City and state
- Latitude and longitude
- Star rating
- Review count
- Business categories
- Business attributes
- Operating hours

The review dataset contains:

- Review ID
- User ID
- Business ID
- Star rating
- Review text
- Review date
- Useful, funny, and cool votes

The business dataset contains approximately 150,346 businesses, while the review dataset contains approximately 6.99 million reviews.

The large size of the dataset makes it suitable for demonstrating data preprocessing and data mining techniques on a real-world dataset.

3 Methodology

The overall methodology of the project is:

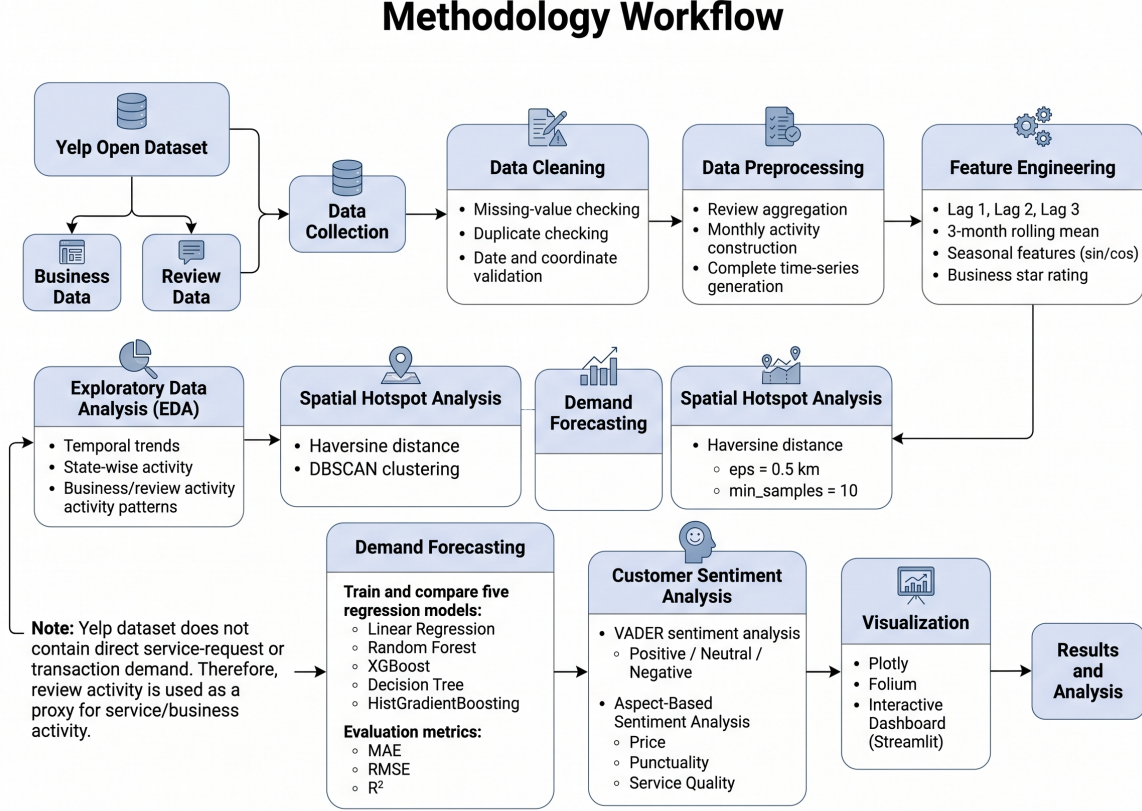


Figure 1: Overall methodology workflow of the proposed spatial-temporal demand and customer sentiment analysis framework.

3.1 Data Preprocessing

The raw Yelp data were stored in JSON format and were processed using Python and Pandas.

The following preprocessing steps were performed:

1. Business and review records were loaded from the Yelp JSON files.
2. Reviews were connected with business information using the common `business_id` field.
3. Review dates were converted into year and month information for temporal analysis.
4. Review activity was aggregated at the business-month level.
5. A complete monthly timeline was created for businesses.
6. Months without reviews were represented as zero activity instead of being removed.
7. Lag features were generated to represent previous monthly activity.

8. A three-month rolling mean was calculated to represent recent historical activity.
9. Seasonal month features were created using sine and cosine transformations.
10. Geographic latitude and longitude values were prepared for spatial clustering.

The final forecasting dataset contained approximately 11.81 million observations and 11 columns.

The forecasting target represents the number of reviews associated with a business in the current month.

3.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed before applying the predictive models. The analysis examined:

- Review activity over different years
- Monthly review activity
- Business review counts
- Business star ratings
- Geographic distribution of businesses
- Review activity across different states

The temporal analysis showed that review activity increased substantially over the earlier years and reached higher levels in the later years of the dataset.

EDA was also useful for identifying the temporal structure required for the forecasting task.

3.3 Spatial Clustering Using DBSCAN

Geographically dense areas of businesses were identified with the use of DBSCAN (Density-Based Spatial Clustering of Applications with Noise).

This clustering was done by using business latitude and longitude coordinates.

The Haversine distance was used, as latitude and longitude were used to represent positions on the surface of the earth.

The final DBSCAN configuration used:

- Distance metric: Haversine
- ϵ : 0.5 km
- Minimum samples: 10

The analysis identified:

- 1,129 spatial clusters
- 136,768 businesses assigned to clusters

- 13,578 businesses identified as noise

For every cluster, the following information was calculated:

- Number of businesses
- Total review count
- Average review count per business
- Average star rating
- Cluster centroid latitude
- Cluster centroid longitude

The DBSCAN analysis provides a spatial perspective of business concentration. It does not directly predict future demand.

3.4 Demand Forecasting Dataset

The forecasting target is the monthly review activity of a business.

The following features were used:

- `stars`
- `lag_1`
- `lag_2`
- `lag_3`
- `rolling_mean_3`
- `month_sin`
- `month_cos`

The lag variables represent historical review activity:

- `lag_1`: previous month's activity
- `lag_2`: activity two months earlier
- `lag_3`: activity three months earlier

The three-month rolling mean represents the recent average activity.

To capture the cyclic nature of months, sine and cosine transformations were used:

$$month_sin = \sin\left(\frac{2\pi m}{12}\right)$$

$$month_cos = \cos\left(\frac{2\pi m}{12}\right)$$

where m is the month number.

3.5 Time-Based Train-Validation-Test Split

A chronological split was used because the forecasting task attempts to predict future activity using historical information.

The dataset was divided as follows:

- Training: 2005–2020
- Validation: 2021
- Test: 2022

Random state 42 was used to select a sample of 1,000,000 training observations that were reproducible.

These data were validated with data from 2021 and the existing 2022 data were used as a held-out test period.

The 2022 test period is a partial test period, as the Yelp review dataset ends on January 19, 2022. Thus, the test results are not to be interpreted as representative of the full year, but rather should be considered as representing performance on the available 2022 observations.

3.6 Forecasting Models

Five applicable regression models were trained and evaluated using the same features and time-based data split.

3.6.1 Random Forest

Random Forest is an ensemble regression method that combines predictions from multiple decision trees.

The implemented configuration was:

- Number of trees = 100
- Maximum depth = 20
- Minimum samples per leaf = 5
- Random state = 42

3.6.2 XGBoost

XGBoost is a gradient boosting algorithm that builds decision trees sequentially and attempts to reduce prediction errors from previous trees.

The implemented configuration was:

- Number of estimators = 300
- Maximum depth = 8
- Learning rate = 0.05
- Subsample = 0.8

- Column sample by tree = 0.8
- Tree method = histogram
- Random state = 42

3.6.3 Linear Regression

Linear Regression was used as a simple baseline regression model.

It estimates the target as a linear combination of the input features.

The model was used to determine how a simple linear relationship compares with more complex nonlinear models.

3.6.4 Decision Tree

Decision Tree Regression recursively divides the feature space into different regions and predicts the target value within each region.

The implemented configuration used:

- Maximum depth = 20
- Minimum samples per leaf = 5
- Random state = 42

3.6.5 HistGradientBoosting

HistGradientBoosting is a gradient boosting regression algorithm that uses histogram-based feature processing.

The implemented configuration used:

- Maximum iterations = 200
- Learning rate = 0.05
- Maximum leaf nodes = 31
- Minimum samples per leaf = 20
- Random state = 42

3.7 Evaluation Metrics

Three standard regression metrics were used to compare the five models.

3.7.1 Mean Absolute Error

Mean Absolute Error (MAE) measures the average absolute difference between actual and predicted values.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Lower MAE indicates smaller average prediction errors.

3.7.2 Root Mean Squared Error

Root Mean Squared Error (RMSE) measures the square root of the average squared prediction error.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

RMSE gives greater influence to large prediction errors.

3.7.3 R^2 Score

The coefficient of determination is given by:

$$R^2 = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

A higher R^2 indicates that more variation in the target is explained by the model.

A negative R^2 indicates that the model performs worse than a baseline that predicts the mean target value.

3.8 Customer Sentiment Analysis

VADER (Valence Aware Dictionary and sEntiment Reasoner) was used to analyze the sentiment of Yelp reviews.

Due to the large size of the review dataset, a reproducible sample of 100,000 reviews was selected using random state 42.

Each review was classified as:

- Positive
- Neutral
- Negative

The sentiment analysis was performed on review text.

To evaluate the sentiment results, Yelp star ratings were converted into proxy sentiment labels:

- 1–2 stars: Negative
- 3 stars: Neutral
- 4–5 stars: Positive

These labels are proxy labels and are not equivalent to human-annotated sentiment labels.

3.9 Aspect-Based Sentiment Analysis

Aspect analysis was performed to identify specific topics discussed in customer reviews.

Three aspects were selected:

- Price
- Punctuality
- Service Quality

A rule-based keyword matching approach was used to identify reviews mentioning each aspect.

After identifying an aspect, the sentiment of the corresponding review was analyzed using the VADER sentiment result.

This allowed the project to examine not only whether a review was positive or negative overall, but also which specific topics were associated with customer opinions.

3.10 Interactive Dashboard

A Streamlit-based interactive dashboard was developed to present the project results.

The dashboard contains five main sections:

1. Overview
2. Spatial Hotspots
3. Demand Forecasting
4. Customer Sentiment
5. Aspect Analysis

The dashboard integrates the outputs of the different analysis stages into one interface.

Users can inspect spatial clusters, forecasting model results, sentiment distributions, and aspect-level sentiment without manually running each analysis script.

4 Results

4.1 Forecasting Model Comparison

The five forecasting models were evaluated using the same chronological training, validation, and test framework.

Table 1 presents the complete results.

Table 1: Comparison of five forecasting models

Model	Val. MAE	Val. RMSE	Val. R^2	Test MAE	Test RMSE	Test R^2
Random Forest	0.6357	1.0872	0.5679	1.0773	1.5858	-0.0867
XGBoost	0.6302	1.0788	0.5746	1.0676	1.5876	-0.0892
Linear Regression	0.6244	1.0841	0.5704	1.0076	1.4242	0.1235
Decision Tree	0.6479	1.1390	0.5258	1.1115	1.7006	-0.2498
HistGradientBoosting	0.6296	1.0715	0.5804	1.0513	1.5332	-0.0159

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4.2 Validation Performance

The validation period corresponds to the year 2021.

HistGradientBoosting produced the highest validation R^2 value of 0.5804 and the lowest validation RMSE of 1.0715.

Its validation MAE was 0.6296.

Linear Regression produced the lowest validation MAE at 0.6244.

The validation results therefore show that different metrics can provide slightly different comparisons between models.

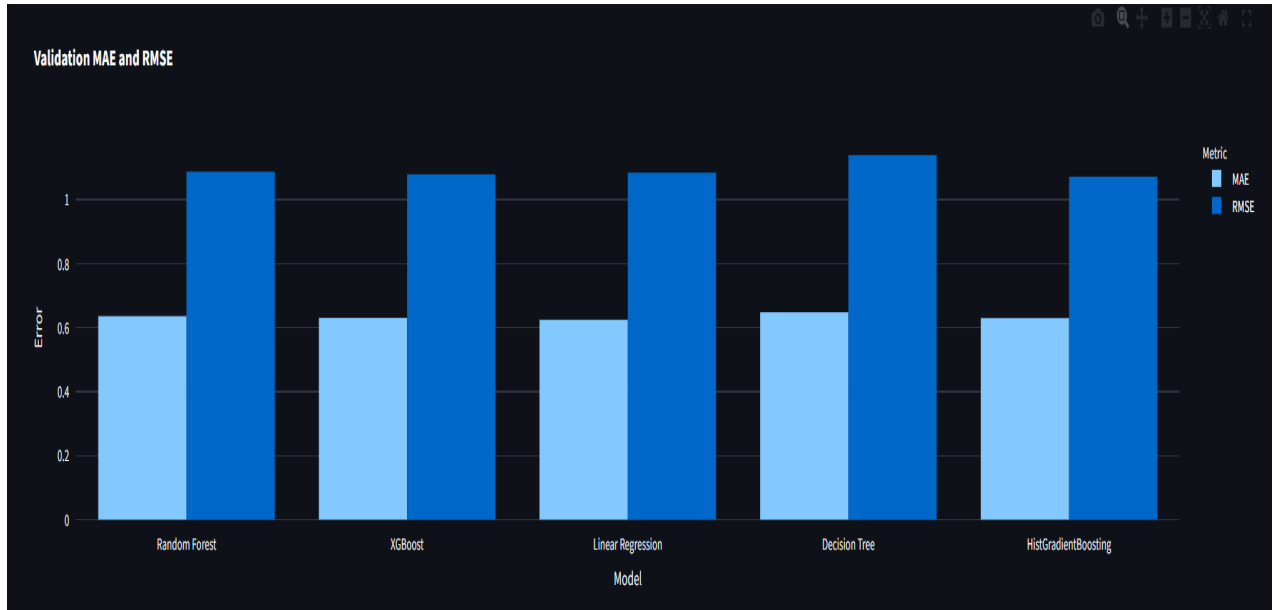


Figure 2: Validation MAE and RMSE comparison across the five forecasting models.

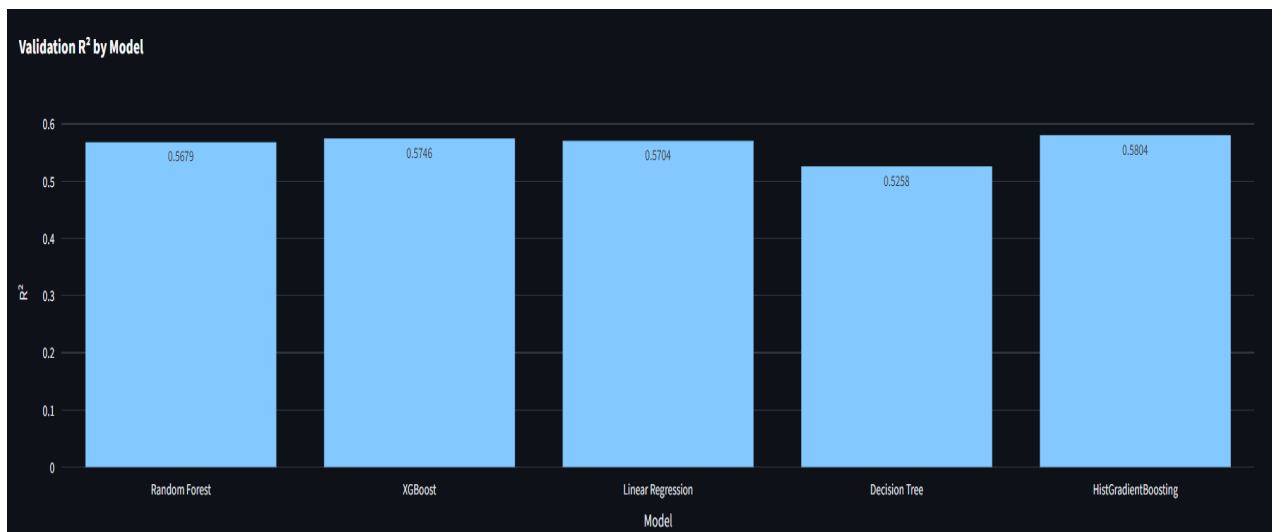


Figure 3: Validation R^2 comparison across the five forecasting models.

4.3 Test Performance

The held-out test period consists of the available 2022 observations.

Linear Regression produced:

- Test MAE = 1.0076
- Test RMSE = 1.4242
- Test $R^2 = 0.1235$

Among the five models, Linear Regression was the only model with a positive test R^2 .

The other four models produced negative test R^2 values.

The results demonstrate that model performance on the validation period did not necessarily remain the same on the unseen 2022 test period.

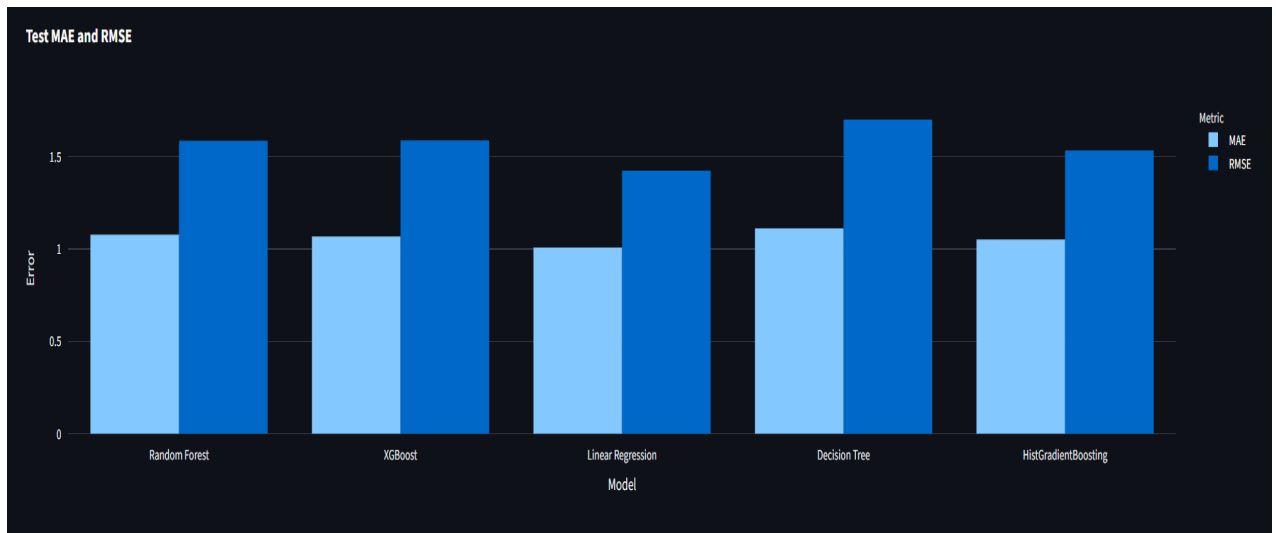


Figure 4: Test MAE and RMSE comparison across the five forecasting models.

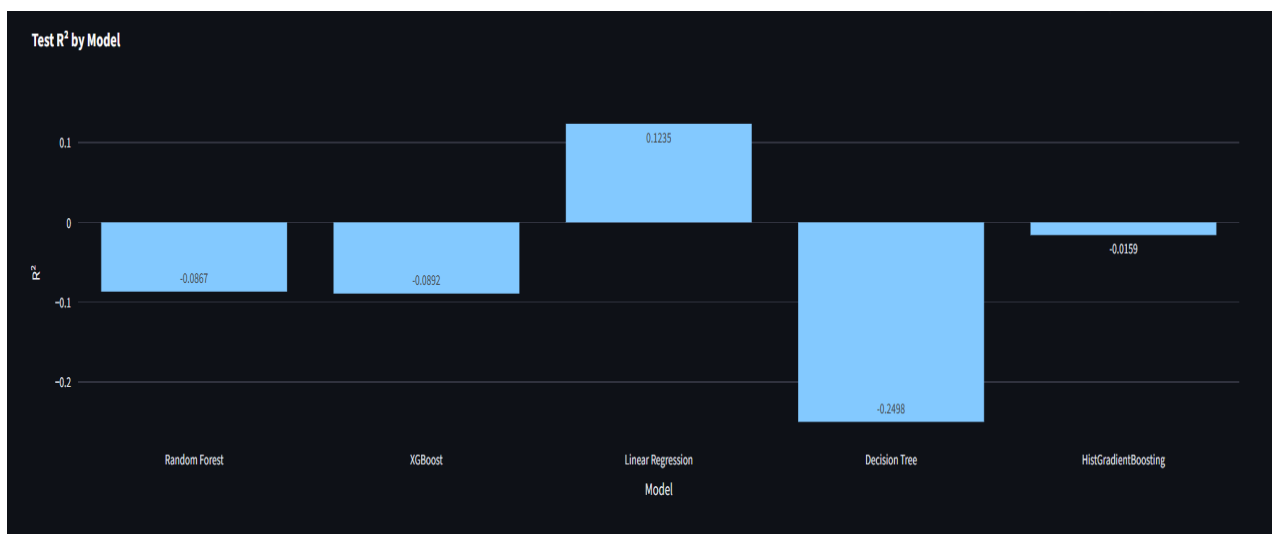


Figure 5: Test R^2 comparison across the five forecasting models.

4.4 Forecasting Results Discussion

The comparison of five regression models demonstrates the importance of using multiple applicable models rather than relying on a single algorithm.

The validation results and test results show different patterns.

The nonlinear boosting models performed strongly on the 2021 validation period. However, the held-out 2022 results were different, with Linear Regression producing the lowest test MAE and RMSE and the only positive test R^2 .

This difference indicates that the models did not generalize equally to the unseen test period.

The forecasting task is challenging because the target represents review activity, which can be sparse and highly variable across businesses.

Therefore, the forecasting results should be interpreted as predictions of review activity rather than direct predictions of restaurant bookings, sales, or service transactions.

4.5 Spatial Hotspot Results

DBSCAN was applied to business locations to identify geographically dense groups of businesses.

Using a Haversine distance of 0.5 km and a minimum of 10 samples, 1,129 clusters were identified.

A total of 136,768 businesses were assigned to clusters, while 13,578 businesses were classified as noise.

The spatial clusters represent areas where businesses are geographically concentrated.

The cluster centroids were visualized using an interactive map.

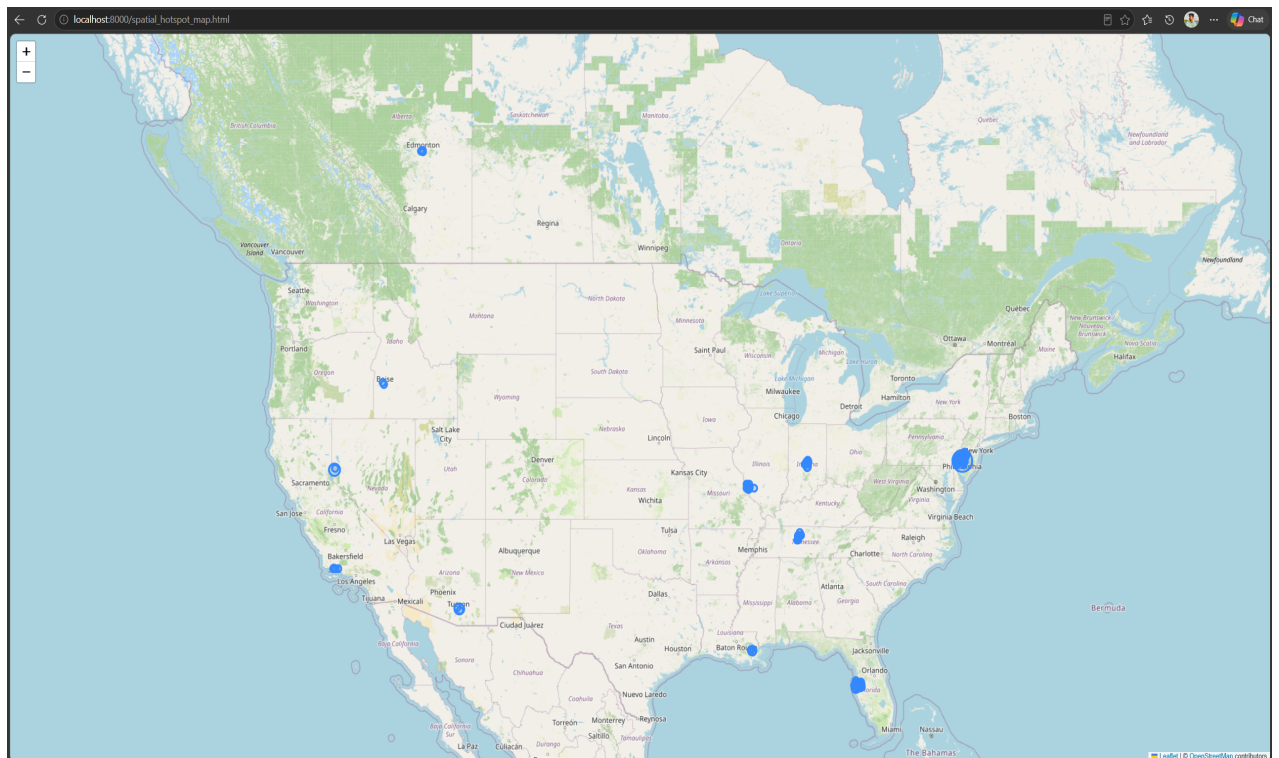


Figure 6: DBSCAN-based spatial hotspot map of Yelp businesses.

The map provides a spatial complement to the forecasting analysis. Forecasting focuses on changes in activity over time, whereas DBSCAN identifies geographic concentrations of businesses.

4.6 Customer Sentiment Results

VADER was applied to a sample of 100,000 Yelp reviews.

The resulting distribution was:

- Positive: 84,321 reviews
- Negative: 14,456 reviews
- Neutral: 1,223 reviews

Thus, most of the sampled reviews were classified as positive by VADER.

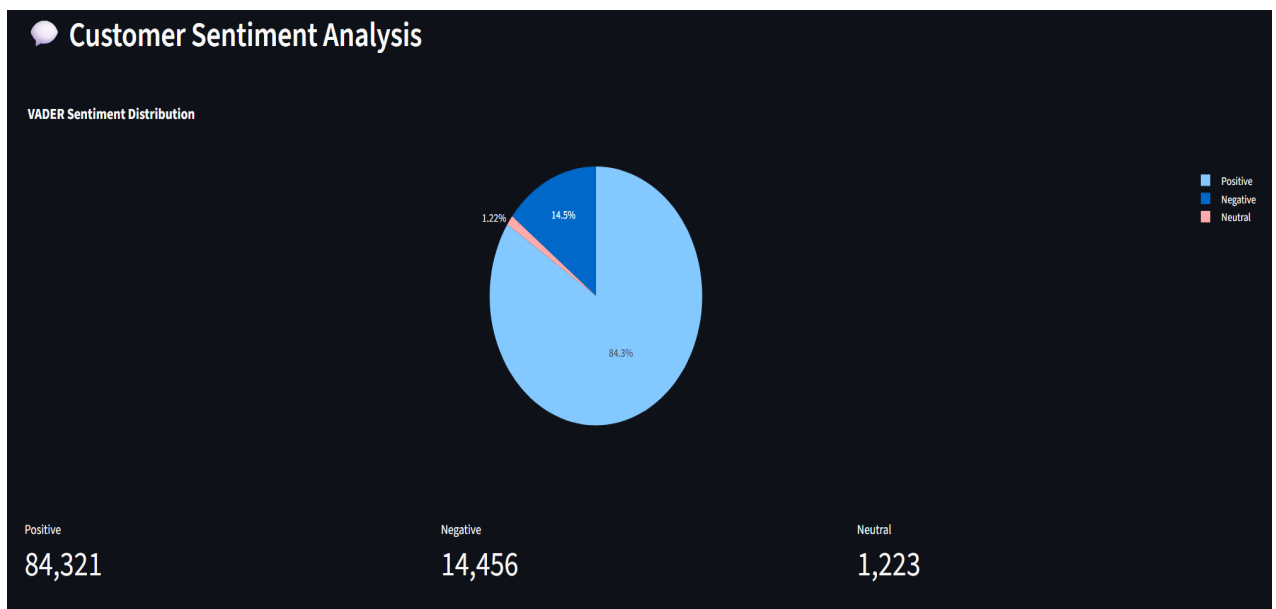


Figure 7: VADER sentiment distribution of the analyzed Yelp review sample.

4.7 Sentiment Evaluation

The VADER sentiment classifications were compared with sentiment labels derived from Yelp star ratings.

The evaluation results were:

Table 2: VADER sentiment evaluation using proxy star-based labels

Metric	Score
Accuracy	0.7777
Weighted Precision	0.7269
Weighted Recall	0.7777
Weighted F1-score	0.7318

The classification results showed that VADER identified positive reviews relatively well, while the Neutral class was difficult to identify.

The Neutral class recall was only 0.02.

Therefore, the overall accuracy should not be interpreted as equally reliable across all three sentiment classes.

The star-based labels are proxy labels rather than human-annotated sentiment labels, which is an important limitation of this evaluation.

4.8 Aspect-Based Sentiment Results

Three aspects were extracted from the reviews:

- Price
- Punctuality
- Service Quality

The sentiment distribution for the extracted aspects is shown in Table 3.

Table 3: Aspect-based sentiment distribution

Aspect	Positive	Neutral	Negative
Price	86.62%	0.86%	12.53%
Punctuality	81.12%	1.13%	17.75%
Service Quality	84.98%	0.72%	14.30%

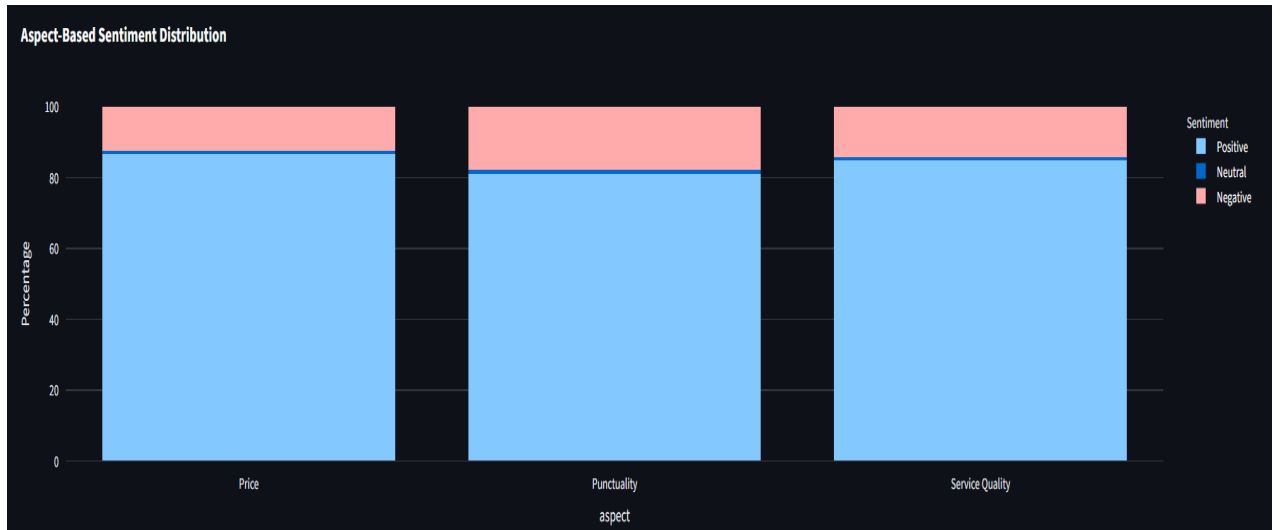


Figure 8: Aspect-based sentiment distribution for Price, Punctuality, and Service Quality.

Among the three extracted aspects, Punctuality had the highest proportion of negative sentiment at 17.75%.

Price had 86.62% positive sentiment, while Service Quality had 84.98% positive sentiment.

The aspect analysis is based on predefined keywords, so implicit or context-dependent mentions may not always be detected.

4.9 Interactive Dashboard

A Streamlit dashboard was developed to integrate the outputs of the complete project. The dashboard includes:

1. Project Overview
2. Spatial Hotspots
3. Demand Forecasting
4. Customer Sentiment
5. Aspect Analysis

The forecasting section displays the comparison of all five regression models using MAE, RMSE, and R^2 .

The spatial section presents the DBSCAN clusters and an interactive map.

The sentiment section presents the VADER sentiment distribution and evaluation results.

The aspect section presents sentiment distributions for Price, Punctuality, and Service Quality.

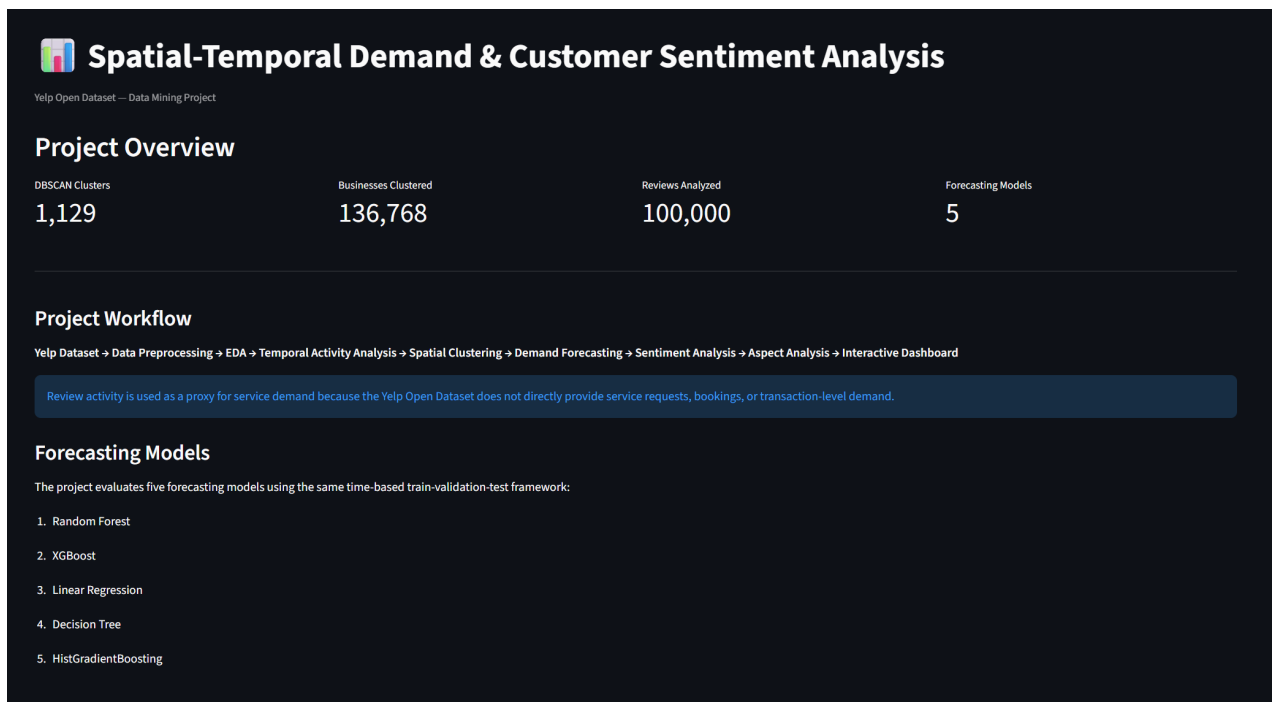


Figure 9: Overview page of the Streamlit interactive dashboard.

5 Discussion

This project puts together multiple Data Mining techniques into one project that analyzes a massive dataset about a real-world in many perspectives.

The preprocessing stage is where raw JSON files were transformed into structured data that can be analyzed.

The fundamental temporal, geographic and business level attributes of data set were comprehended with EDA.

DBSCAN gave a spatial orientation with the identification of geographic clusters of businesses.

The forecasting component was the main predictive component of the project. The same feature set and train/validation/test split was used to train five regression models. The three metrics of MAE, RMSE and R^2 were used to compare their results.

The outcome of the forecasting suggested there would be a difference between the validation period and the test period. This is a good reminder of the need to test several models and hold out a future time period.

The result of the sentiment analysis part provided a more general information regarding the customers' voice as shown in the review texts, and the result of aspect analysis part provided more specific information about the customers' voice on Price, Punctuality and Service Quality aspects.

Finally, the different results of the analysis were put into one interactive dashboard in the Streamlit dashboard.

6 Limitations

The project has several limitations that should be considered when interpreting the results.

1. The Yelp Open Dataset does not contain direct service requests, bookings, sales, or transaction-level demand. Therefore, review activity is used as a proxy for service demand.
2. The forecasting models predict monthly review activity rather than actual customer bookings or transactions.
3. The forecasting models were trained using a reproducible 1,000,000-row sample from the available training observations.
4. The 2022 test period is partial because the available review data end on January 19, 2022.
5. The sentiment evaluation uses Yelp star ratings as proxy labels rather than human-annotated sentiment labels.
6. VADER showed weak performance for the Neutral sentiment class.
7. Aspect extraction uses predefined keywords and therefore may fail to identify implicit or context-dependent aspect mentions.
8. DBSCAN results depend on parameters such as the distance threshold and minimum number of samples.
9. Review date does not necessarily represent the actual date when a customer received a service. Therefore, temporal review activity should not be interpreted as exact transaction-level service demand.

7 Conclusion

This project presents a complete data mining workflow using the Yelp Open Dataset.

The raw business and review data were processed and transformed into structured spatial and temporal datasets.

Exploratory data analysis was performed to understand the characteristics of business and review activity.

DBSCAN was used to identify geographic concentrations of businesses.

For the main forecasting task, five regression models were trained and compared:

- Random Forest
- XGBoost
- Linear Regression
- Decision Tree
- HistGradientBoosting

The models were evaluated using MAE, RMSE, and R^2 on validation and held-out test data.

HistGradientBoosting produced strong validation results, while Linear Regression produced the lowest test MAE and RMSE and the only positive test R^2 among the five models.

The project also included VADER sentiment analysis on 100,000 sampled reviews. Most of the analyzed reviews were classified as positive.

Aspect-based analysis was then performed for Price, Punctuality, and Service Quality. Punctuality showed the highest proportion of negative sentiment among the three selected aspects.

The final Streamlit dashboard integrates the spatial, forecasting, sentiment, and aspect analysis results.

Overall, the project demonstrates how preprocessing, exploratory analysis, spatial clustering, multiple machine learning models, model comparison, sentiment analysis, and visualization can be combined into a single data mining workflow for large-scale review data.

8 Code Link

The complete source code and project files will be made publicly available through GitHub.

<https://github.com/YOUR-USERNAME/YOUR-REPOSITORY>

The repository will contain:

- Data preprocessing scripts
- EDA scripts
- DBSCAN clustering implementation

- Forecasting model implementations
- Five-model comparison results
- Sentiment analysis
- Aspect analysis
- Streamlit dashboard

9 References

1. Yelp Dataset, Kaggle. <https://www.kaggle.com/yelp-dataset/yelp-dataset>
2. Yelp Developer Documentation. <https://docs.developer.yelp.com/>
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4. Chen, T. and Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining.